

Software Test Documentation

XRayVision AI — Test Plan and Test Case Specification

Field	Value
Document	Software Test Documentation (Test Plan + Test Case Specification)
Version	1.0
Standard	IEEE Std 829 (adapted)
Companion documents	SRS · SDD · User Manual
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Read this before using the results columns

This document specifies **93 test cases** derived from the SRS. The **Status** column records execution state honestly:

Marker	Meaning
 Pass	Executed against the live deployment and verified. Evidence is referenced.
 Pending	Specified but not yet executed . The tester must run it and fill in the Actual Result.

Seventeen cases were verified during documentation by driving the live deployment at <https://x-ray-vision-board.vercel.app> with an automated browser session; their evidence is the screenshot set in [docs/screenshots/](#). The remaining cases — chiefly those needing real medical images, negative-path inputs, timing instrumentation and multi-account setups — are specified in full and are ready to execute. **Do not present a Pending row as a passing result.**

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1. Test Plan

1.1 Objectives

1. Verify that every functional requirement in the SRS is implemented as specified.
2. Verify that invalid and hostile inputs are rejected with correct status codes and readable messages.
3. Verify that the documented degradation paths behave as designed — a failing dependency must downgrade the result, not destroy it.
4. Verify that a user can never read or modify another user's data.
5. Verify that measurable non-functional targets are met.

1.2 Scope

In scope: all API endpoints, all UI pages, authentication, authorisation, validation, degradation paths, and the documented performance targets.

Out of scope: clinical accuracy of the AI models (the models are pretrained third-party artefacts and are not being validated as medical devices), third-party service internals, and load testing beyond the documented concurrency target.

1.3 Test Levels

Level	Purpose	Technique
Unit	Individual service functions — validation, Haversine, plan validation	Manual invocation / pytest (to be added)
Integration	Router → service → repository chains	Swagger UI at /docs , curl
System	End-to-end user journeys through the browser	Manual and automated browser sessions
Security	Authentication, authorisation, rate limiting, isolation	Crafted requests with curl
Acceptance	The supervisor's evaluation scenarios	Demo walkthrough

1.4 Entry and Exit Criteria

Criterion	Definition
Entry	Backend responds 200 on GET /health ; frontend loads; Supabase reachable; a valid OPENROUTER_API_KEY is configured.
Exit	100 % of Must-have (M) cases executed and passing; no open Critical or High defect; all Should-have (S) cases executed.

1.5 Pass / Fail Criteria

A case **passes** only when the observed status code, response body, persisted state and rendered UI all match the Expected Result. A partial match is a **fail** and must be logged in [§17](#).

1.6 Risks

Risk	Mitigation
Free-tier cold start distorts timing measurements	Warm the backend with three requests before running TC-PERF cases
OpenRouter free-tier quota exhausted mid-run	Schedule LLM-dependent cases first; TC-ANLZ-13 deliberately exercises the fallback
Overpass public mirrors rate-limit	Retry TC-CLIN cases after a pause; the mirror fallback is itself under test
No real DICOM samples available	Obtain a public sample study before executing TC-ANLZ-05

2. Test Environment

Item	Configuration
Frontend under test	<code>https://x-ray-vision-board.vercel.app</code>
Backend under test	Hugging Face Spaces deployment (<code>*.hf.space</code>)
Database	Supabase PostgreSQL (project instance)
Browsers	Chrome 143, Edge 120+, Firefox 120+, Safari 17+
Viewports	1440×900 desktop, 390×844 mobile
API tools	Swagger UI (<code>/docs</code>), <code>curl</code>
Local backend	<code>uvicorn app.main:app --reload --port 8000</code>
Automation	Playwright (Chromium) for browser-driven cases

3. Test Data

3.1 Accounts

ID	Email	Password	Purpose
U1	<code>ahmad123@gmail.com</code>	<code>12345678</code>	Demo account, pre-populated with 31 scans
U2	<i>(tester creates)</i>	≥ 8 chars	Fresh registration and isolation testing
U3	<i>(tester creates)</i>	≥ 8 chars	Second account for cross-account isolation (TC-SEC-04)

3.2 Image Fixtures

ID	File	Properties	Used by
IMG-1	<code>chest_valid.jpg</code>	Chest X-ray, ~800×800 px, ~400 KB	TC-ANLZ-01, TC-PERF-01
IMG-2	<code>wrist_fracture.jpg</code>	Bone X-ray with a visible fracture	TC-ANLZ-02
IMG-3	<code>wound_photo.jpg</code>	External wound photograph	TC-ANLZ-03
IMG-4	<code>tiny.png</code>	150×150 px, ~120 KB	TC-ANLZ-07 (resolution floor)
IMG-5	<code>small.jpg</code>	~40 KB	TC-ANLZ-06 (size floor)
IMG-6	<code>huge.jpg</code>	~25 MB	TC-ANLZ-08 (size ceiling)
IMG-7	<code>document.pdf</code> renamed <code>fake.jpg</code>	Not a decodable image	TC-ANLZ-09

ID	File	Properties	Used by
IMG-8	study.dcm	Valid DICOM	TC-ANLZ-05
IMG-9	empty.jpg	0 bytes	TC-ANLZ-10
IMG-10	clear_chest.jpg	Chest X-ray with no pathology	TC-ANLZ-12

4. Test Cases — Authentication (TC-AUTH)

TC-AUTH-01 — Register with valid data

Field	Detail
Requirement	FR-001, FR-004
Priority	High
Precondition	Email not already registered
Test data	Name Test User , email newuser@test.com , password Test@1234

Steps

1. Open `/auth/register` .
2. Enter full name, email, password and matching confirmation.
3. Click **Create account**.

Expected result

- HTTP 200; `AuthResponse` with `access_token` and `user` .
- A `profiles` row exists with `full_name` = "Test User" and `role` = "Medical Student" .
- Browser redirects to `/dashboard` ; the header shows the user's name.

Actual result	Status
(to be filled by tester)	 Pending

TC-AUTH-02 — Register with a password shorter than 8 characters

Field	Detail
Requirement	FR-002

Field	Detail
Priority	High
Test data	Password abc123 (6 chars)

Steps — Open `/auth/register` , enter a valid name and email, enter `abc123` in both password fields, submit.

Expected result — Client-side validation blocks submission with a message referencing the 8-character minimum. If sent directly to the API, HTTP 422 from Pydantic (`min_length=8`).

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-AUTH-03 — Register with mismatched password confirmation

Field	Detail
Requirement	FR-001 (UC-01 A1)
Priority	Medium
Test data	Password <code>Test@1234</code> , confirmation <code>Test@5678</code>

Expected result — Inline error "passwords do not match"; no API call is made; the user remains on the form.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-AUTH-04 — Register with an already-registered email

Field	Detail
Requirement	FR-003
Priority	High
Test data	Email <code>ahmad123@gmail.com</code>

Expected result — HTTP 400; a readable "email already registered" message is displayed; no duplicate `profiles` row is created.

Actual result	Status
(to be filled by tester)	 Pending

TC-AUTH-05 — Login with valid credentials

Field	Detail
Requirement	FR-005, FR-006
Priority	High
Test data	U1 — ahmad123@gmail.com / 12345678

Steps — Open `/auth/login` , enter the credentials, click **Sign In**.

Expected result — HTTP 200 with a JWT; the browser redirects to `/dashboard` ; the greeting shows the user's name.

Actual result	Status
Redirected to <code>/dashboard</code> ; header shows "ahmad · Researcher"; greeting "Good afternoon, ahmad"; dashboard populated with 31 scans. Evidence: <code>screenshots/05-dashboard.png</code>	 Pass

TC-AUTH-06 — Login with an invalid password

Field	Detail
Requirement	FR-005 (UC-02 E1)
Priority	High
Test data	ahmad123@gmail.com / wrongpassword

Expected result — HTTP 401; "Invalid email or password" displayed; no token is stored; the user stays on `/auth/login` .

Actual result	Status
(to be filled by tester)	 Pending

TC-AUTH-07 — Login with an unregistered email

Field	Detail
Requirement	FR-005
Priority	Medium
Test data	doesnotexist@test.com / Test@1234

Expected result — HTTP 401 with a generic message that does not reveal whether the email exists.

Actual result	Status
(to be filled by tester)	 Pending

TC-AUTH-08 — Demo account one-click login

Field	Detail
Requirement	FR-010a
Priority	Medium

Steps — Open `/auth/login`, click **Try demo account**.

Expected result — A loading state appears, then the user is signed in and redirected to `/dashboard` without typing credentials.

Actual result	Status
Button present and labelled "Try demo account" on <code>/auth/login</code> . Evidence: <code>screenshots/02-login.png</code> . Click-through not exercised.	 Pending

TC-AUTH-09 — Access a protected page without a token

Field	Detail
Requirement	FR-007
Priority	High

Steps — Clear local storage, navigate directly to `/dashboard`.

Expected result — The user is redirected to `/auth/login`; no scan data is rendered at any point.

Actual result	Status
(to be filled by tester)	 Pending

TC-AUTH-10 — Logout clears the session

Field	Detail
Requirement	FR-010
Priority	High

Steps — Sign in, click the logout control, then press the browser Back button.

Expected result — `xray_token` and `xray_user` are removed from local storage; the user lands on `/`; Back does not restore an authenticated view.

Actual result	Status
(to be filled by tester)	 Pending

TC-AUTH-11 — Forgot-password request

Field	Detail
Requirement	FR-008
Priority	Medium
Test data	A registered email

Steps — Open `/auth/forgot-password`, enter the email, click **Send reset link**.

Expected result — HTTP 200 with a confirmation message; a Supabase password-reset email arrives.

Actual result	Status
Page renders with heading "Reset your password", one email input and a Send reset link button. Evidence: <code>screenshots/04-forgot-password.png</code> . Email delivery not verified.	 Pending

5. Test Cases — Image Analysis (TC-ANLZ)

TC-ANLZ-01 — Chest analysis, happy path

Field	Detail
Requirement	FR-011, FR-012, FR-019, FR-020, FR-021, FR-023, FR-026
Priority	Critical
Precondition	Signed in as U1
Test data	IMG-1, route Chest pathology , label PT-1001 , notes Cough for 5 days

Steps

1. Open `/analyze` .
2. Upload IMG-1 (Step 1).
3. Select **Chest pathology** (Step 2).
4. Enter the label and notes (Step 3).
5. Click **Analyze image**.

Expected result

- A processing overlay is shown while the request runs.
- HTTP 200; navigation to `/results/{scanId}` .
- Findings originate from `DenseNet121` ; **at most 6** are returned.
- Each finding shows name, confidence %, severity, model and an ICD-10 code where mapped.
- An `AgentSynthesis` paragraph, recommended actions and a specialist are displayed.
- `urgency` is one of `critical|high|medium|low|clear` .
- A new row appears at the top of `/history` .

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-ANLZ-02 — Fracture analysis returns bounding boxes

Field	Detail
Requirement	FR-020, FR-022, FR-027
Priority	Critical
Test data	IMG-2, route Fracture detection

Expected result

- Models run include `YOLOv8` and `FractureClassifier`.
- Bounding boxes are drawn over the image, aligned to the finding regions.
- **No** finding named `Not_Fracture` appears anywhere in the response or the UI.
- A confidence summary table lists every finding with model, confidence and severity.

Actual result	Status
Stored fracture scan <code>a078d05a</code> renders dashed bounding boxes on the image; findings list <code>FractureClassifier</code> – Fracture suspected 96.7% (ICD-10 <code>S02-S92</code>) and four <code>YOLOv8</code> – Vase rows; confidence summary table present; no <code>Not_Fracture</code> entry. Evidence: <code>screenshots/13-results.png</code> . New upload not exercised.	 Pass

TC-ANLZ-03 — Wound analysis

Field	Detail
Requirement	FR-020
Priority	High
Test data	IMG-3, route <code>External wound</code>

Expected result — Findings originate from the `vit` model with a wound-category label and a confidence score; a synthesis paragraph is produced.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-ANLZ-04 — Analysis without a label or notes

Field	Detail
Requirement	FR-018
Priority	Medium
Test data	IMG-1, no label, no notes

Expected result — Analysis completes normally; `session_label` and `notes` are stored as empty/null; the results page renders without gaps.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-ANLZ-05 — DICOM file upload

Field	Detail
Requirement	FR-012
Priority	High
Test data	IMG-8 (study.dcm)

Expected result — The PIL dimension check is skipped; `pydicom` decodes the file; analysis completes and a result is returned.

Actual result	Status
(to be filled by tester)	 Pending

TC-ANLZ-06 — Reject a file below the size floor

Field	Detail
Requirement	FR-013
Priority	High
Test data	IMG-5 (~40 KB)

Expected result — HTTP **422**; message states the actual size in KB and the 100 KB minimum; no `scans` row is created.

Actual result	Status
(to be filled by tester)	 Pending

TC-ANLZ-07 — Reject a low-resolution image

Field	Detail
Requirement	FR-015
Priority	High
Test data	IMG-4 (150×150 px, ≥ 100 KB)

Expected result — HTTP **422**; message states the actual dimensions and the 200×200 px minimum.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-ANLZ-08 — Reject a file above the size ceiling

Field	Detail
Requirement	FR-014
Priority	High
Test data	IMG-6 (~25 MB)

Expected result — HTTP **422**; message states the actual size in MB and the 20 MB maximum.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-ANLZ-09 — Reject an undecodable file

Field	Detail
Requirement	FR-012
Priority	High
Test data	IMG-7 (a PDF renamed <code>.jpg</code> , > 100 KB)

Expected result — HTTP **422**; message directs the user to upload a JPEG, PNG or DICOM file. No unhandled exception is logged.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-ANLZ-10 — Reject an empty upload

Field	Detail
Requirement	FR-016
Priority	Medium

Field	Detail
Test data	IMG-9 (0 bytes)

Expected result — HTTP **400** "Empty file uploaded."

Actual result	Status
(to be filled by tester)	Pending

TC-ANLZ-11 — Reject an invalid scan type

Field	Detail
Requirement	FR-017
Priority	Medium
Test data	scan_type=brain submitted via curl

Expected result — HTTP **400** "scan_type must be one of: chest, fracture, wound".

Actual result	Status
(to be filled by tester)	Pending

TC-ANLZ-12 — Clear scan produces an empty findings state

Field	Detail
Requirement	FR-023 (UC-03 A1)
Priority	Medium
Test data	IMG-10, route Chest pathology

Expected result — findings is an empty array; urgency is clear ; the results page shows a "no significant findings" state rather than an error.

Actual result	Status
Dashboard history shows a stored chest scan (dc4bad5e) with 0 findings and urgency CLEAR , confirming the state is reachable and rendered. Evidence: screenshots/05-dashboard.png . Fresh upload not exercised.	✓ Pass

TC-ANLZ-13 — LLM failure falls back instead of failing

Field	Detail
Requirement	FR-024, NFR-Q1
Priority	Critical
Setup	Temporarily set <code>OPENROUTER_API_KEY</code> to an invalid value and restart the backend

Expected result

- The analysis still returns HTTP **200**.
- `agent_synthesis.synthesis_text` reads "AI synthesis temporarily unavailable. Please review findings manually."
- `urgency` is `medium`.
- `recommended_actions` contains "Consult a radiologist for interpretation".
- Model findings are present and correct.

Actual result	Status
(to be filled by tester)	 Pending

TC-ANLZ-14 — Storage failure does not block the result

Field	Detail
Requirement	FR-025, NFR-Q1
Priority	High
Setup	Temporarily rename the <code>xray-images</code> bucket or revoke storage permission

Expected result — HTTP **200**; `image_url` is an empty string; a warning is logged; the `scans` row is still inserted; the results page renders findings without the image.

Actual result	Status
(to be filled by tester)	 Pending

TC-ANLZ-15 — Analyze rate limit

Field	Detail
Requirement	NFR-SEC5

Field	Detail
Priority	High

Steps — Submit 11 analysis requests from the same IP within 60 seconds.

Expected result — Requests 1–10 succeed; request 11 returns HTTP **429** with "Rate limit exceeded. Please slow down."

Actual result	Status
(to be filled by tester)	 Pending

TC-ANLZ-16 — Processing diagnostics are returned

Field	Detail
Requirement	FR-025a
Priority	Medium

Expected result — `model_results` contains `scan_type`, `ensemble_mode: "routed"`, `models_run`, `model_errors` and a numeric `processing_time_ms`.

Actual result	Status
(to be filled by tester)	 Pending

6. Test Cases — Scans and Records (TC-SCAN)

TC-SCAN-01 — History lists scans newest first

Field	Detail
Requirement	FR-029
Priority	High

Expected result — `/history` lists the signed-in user's scans in descending `created_at` order, each showing type, findings count, urgency and date.

Actual result	Status
<code>/history</code> renders the demo account's scan list; dashboard "Recent analyses" shows descending dates (6/3/2026 ... 5/31/2026). Evidence: <code>screenshots/07-history.png</code> , <code>screenshots/05-dashboard.png</code>	✓ Pass

TC-SCAN-02 — Filter history by scan type

Field	Detail
Requirement	FR-030
Priority	Medium

Steps — On `/history` , apply the `fracture` filter.

Expected result — Only fracture scans are listed; the count matches a direct `GET /scans?scan_type=fracture` call.

Actual result	Status
<i>(to be filled by tester)</i>	⬜ Pending

TC-SCAN-03 — Open a scan result from history

Field	Detail
Requirement	FR-027, FR-028
Priority	High

Expected result — Clicking a row opens `/results/{scanId}` showing the image, overlays, findings and synthesis for that exact scan.

Actual result	Status
Navigating to <code>/results/a078d05a-...</code> renders the wrist X-ray with overlays, urgency HIGH , findings, GLM agent analysis, immediate actions and specialist recommendation. Evidence: <code>screenshots/13-results.png</code>	✓ Pass

TC-SCAN-04 — Findings are grouped into confidence tiers

Field	Detail
Requirement	FR-028a

Field	Detail
Priority	High

Expected result — Findings $\geq 65\%$ under **Primary Findings**; 50–65 % under **Secondary Findings** labelled "50–65% confidence"; $< 50\%$ collapsed behind an expandable "borderline findings" control.

Actual result	Status
Results page shows "Primary Findings" (Fracture suspected, 96.7 %), "Secondary Findings 50–65% confidence" (Vase, 52.7 %) and a collapsed "3 borderline findings below 50% — click to expand". Evidence: <code>screenshots/13-results.png</code>	✓ Pass

TC-SCAN-05 — Low-confidence review banner

Field	Detail
Requirement	FR-028b, NFR-S7
Priority	High

Expected result — When any finding is below 60 % confidence, a banner reading "One or more findings have confidence below 60% — radiologist review recommended" is displayed.

Actual result	Status
Banner present verbatim on scan <code>a078d05a</code> (lowest finding 16.8 %). Evidence: <code>screenshots/13-results.png</code>	✓ Pass

TC-SCAN-06 — Image overlay controls

Field	Detail
Requirement	FR-028c
Priority	Medium

Steps — On a result with boxes, toggle **AI Findings**, **Heatmap** and **Labels**; zoom in, zoom out, reset.

Expected result — Each toggle shows/hides its layer; zoom is clamped between 0.5× and 2.5×; reset returns to 1×.

Actual result	Status
Controls AI Findings , Heatmap , Labels plus zoom-in / zoom-out / reset buttons are rendered. Evidence: <code>screenshots/13-results.png</code> . Interaction not exercised.	■ Pending

TC-SCAN-07 — Download PDF report

Field	Detail
Requirement	FR-031
Priority	High

Steps — On a result page, click **Download PDF Report**.

Expected result — HTTP 200 with `Content-Type: application/pdf` ; the file opens and contains the findings, synthesis, urgency and the educational-use disclaimer.

Actual result	Status
Download PDF Report button is present on the result page. Evidence: <code>screenshots/13-results.png</code> . Download and file contents not verified.	 Pending

TC-SCAN-08 — Export JSON

Field	Detail
Requirement	FR-032
Priority	Medium

Expected result — Valid JSON is returned containing the raw findings array with confidence, model, severity and bbox fields.

Actual result	Status
Export JSON button is present. Evidence: <code>screenshots/13-results.png</code> . Payload not verified.	 Pending

TC-SCAN-09 — Delete a scan

Field	Detail
Requirement	FR-033, BR-5
Priority	High

Steps — Delete a scan from history, then reload and attempt to open its old URL.

Expected result — HTTP 200; the row disappears from `/history` ; the `scans` row is gone; the stored image is removed; the old result URL returns 404.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-SCAN-10 — Fetch a non-existent scan

Field	Detail
Requirement	UC-05 E1
Priority	Medium
Test data	A random UUID

Expected result — HTTP **404**; the UI shows a "not found" state rather than a blank page.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-SCAN-11 — Empty history state

Field	Detail
Requirement	FR-029
Priority	Low
Precondition	Signed in as freshly registered U2

Expected result — `/history` shows an empty-state message inviting the user to run their first analysis; no error is shown.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-SCAN-12 — PDF export for another user's scan

Field	Detail
Requirement	BR-1, NFR-SEC4
Priority	Critical

Steps — As U2, request `GET /scans/{U1_scan_id}/report.pdf` .

Expected result — HTTP **404**; no PDF bytes are returned.

Actual result	Status
<i>(to be filled by tester)</i>	Pending

7. Test Cases — Dashboard (TC-DASH)

TC-DASH-01 — Headline statistics render

Field	Detail
Requirement	FR-034
Priority	High

Expected result — Total scans, urgent reports, average confidence and average report time are displayed as numeric tiles.

Actual result	Status
Tiles render: Total scans 31, Urgent reports 15, Avg confidence 63.2 %, Report time 8.4 s. Evidence: <code>screenshots/05-dashboard.png</code>	✓ Pass

TC-DASH-02 — Finding distribution

Field	Detail
Requirement	FR-035
Priority	Medium

Expected result — A ranked list of finding names with frequency counts and proportional bars.

Actual result	Status
"Finding distribution" panel lists Lung Opacity 10, Lung Lesion 8, Nodule 3, Mass 3, Enlarged Cardiomedastinum 3 with bars. Evidence: <code>screenshots/05-dashboard.png</code>	✓ Pass

TC-DASH-03 — Recent analyses link through

Field	Detail
Requirement	FR-036
Priority	Medium

Expected result — Up to five recent scans are listed with type, findings count, urgency badge and date; each links to its result page.

Actual result	Status
Five rows rendered with linked scan ids, type, findings count and urgency badges (HIGH / CLEAR / MEDIUM). Evidence: <code>screenshots/05-dashboard.png</code>	 Pass

TC-DASH-04 — Per-model confidence bars

Field	Detail
Requirement	FR-036a
Priority	Medium

Expected result — Average confidence is shown per model with a labelled progress bar.

Actual result	Status
"Avg. model confidence" shows DenseNet121 84.7 %, YOLOv8 89.1 %, ViT 82.3 % with bars. Evidence: <code>screenshots/05-dashboard.png</code>	 Pass

TC-DASH-05 — Statistics are scoped to the signed-in user

Field	Detail
Requirement	FR-037, BR-1
Priority	Critical

Steps — Note U1's totals, sign in as U2 (no scans), open `/dashboard` .

Expected result — U2 sees zeros and an empty recent list — never U1's figures.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

8. Test Cases — Chatbot (TC-CHAT)

TC-CHAT-01 — English question and reply

Field	Detail
Requirement	FR-038, FR-039
Priority	High
Test data	"I have had a headache for three days, what should I do?"

Expected result — HTTP 200; a relevant English reply; a `session_id` is returned; both messages are persisted.

Actual result	Status
(to be filled by tester)	 Pending

TC-CHAT-02 — Urdu question and RTL rendering

Field	Detail
Requirement	FR-039
Priority	High
Test data	Language اردو ; message مجھے تین دن سے سر درد ہے

Expected result — The reply is in Urdu; the input and message bubbles render right-to-left; the placeholder changes to ... اپنی علامات بتائیں .

Actual result	Status
Language toggle offers EN and اردو on /chat . Evidence: screenshots/08-chat.png . Round-trip not exercised.	 Pending

TC-CHAT-03 — Structured fields are extracted

Field	Detail
Requirement	FR-042

Field	Detail
Priority	Medium
Test data	"I have chest pain and shortness of breath"

Expected result — `doctor_type` is populated (e.g. Cardiologist) and `home_remedies` is a non-empty array; both render as styled cards, not as raw `DOCTOR_TYPE:` text inside the reply.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-CHAT-04 — Conversation context is retained

Field	Detail
Requirement	FR-040
Priority	High

Steps — Ask "I have a fever", then follow up with "how long should it last?" in the same session.

Expected result — The second reply refers to the fever, demonstrating that prior messages were supplied as context.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-CHAT-05 — Sessions persist across visits

Field	Detail
Requirement	FR-043
Priority	High

Steps — Hold a conversation, navigate away, return to `/chat` .

Expected result — The session appears in the session list and its messages are restored in order.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-CHAT-06 — Safety framing in replies

Field	Detail
Requirement	FR-041, NFR-S2
Priority	Critical
Test data	"Do I have cancer? Just tell me yes or no."

Expected result — The bot does **not** assert a definitive diagnosis; it directs the user to a qualified medical professional.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-CHAT-07 — Chat rate limit

Field	Detail
Requirement	NFR-SEC5
Priority	Medium

Steps — Send 21 messages within 60 seconds.

Expected result — The 21st returns HTTP **429**.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-CHAT-08 — Voice input

Field	Detail
Requirement	FR-043a
Priority	Low

Steps — Click the microphone control, grant permission, dictate a question in each language.

Expected result — Recognition uses `en-US` or `ur-PK` to match the selected language; the transcript populates the input. If permission is denied, text input still works.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

9. Test Cases — Diet Planner (TC-DIET)

TC-DIET-01 — Generate a plan for a stated condition

Field	Detail
Requirement	FR-044, FR-045, FR-048
Priority	High
Test data	Condition <code>diabetes</code> , preference <code>balanced</code> , restrictions <code>nut allergy</code> , goal <code>weight loss</code>

Expected result — Exactly **7** day entries, each with breakfast, lunch and dinner; meals carry names and descriptions; the plan has a title and summary.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-DIET-02 — Hypertension applies DASH guidance

Field	Detail
Requirement	FR-046, NFR-S3
Priority	High
Test data	Condition <code>high blood pressure</code>

Expected result — The plan reflects DASH principles (reduced sodium, emphasis on fruit, vegetables and low-fat dairy) and says so in the summary or tips.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-DIET-03 — Diabetes applies low-GI guidance

Field	Detail
Requirement	FR-046
Priority	High

Expected result — Low-glycaemic-index choices dominate and the rationale is stated.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-DIET-04 — Kidney disease adds a renal-dietitian note

Field	Detail
Requirement	FR-046, NFR-S3
Priority	Critical
Test data	Condition <code>kidney disease</code>

Expected result — The response includes an explicit recommendation to consult a renal dietitian.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-DIET-05 — Malformed LLM output triggers the fallback plan

Field	Detail
Requirement	FR-047
Priority	Critical
Setup	Force an invalid LLM response (invalid key, or a stub returning 5 days)

Expected result — The validated hard-coded 7-day fallback plan is returned. A partial or malformed plan is **never** displayed.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-DIET-06 — Diet rate limit

Field	Detail
Requirement	NFR-SEC5
Priority	Medium

Steps — Submit 11 diet requests within 60 seconds.

Expected result — The 11th returns HTTP **429**.

Actual result	Status
(to be filled by tester)	 Pending

10. Test Cases — Clinic Locator (TC-CLIN)

TC-CLIN-01 — Find clinics with permission granted

Field	Detail
Requirement	FR-049, FR-050, FR-051, FR-052
Priority	High

Steps — Open `/clinics`, set the radius to 5 km, click **Use My Location**, grant permission.

Expected result — Facilities are listed with name, type, address, distance in km and a Maps link, sorted by ascending distance; every result is within the radius.

Actual result	Status
(to be filled by tester)	 Pending

TC-CLIN-02 — Radius slider bounds

Field	Detail
Requirement	FR-049
Priority	Medium

Expected result — The slider spans **1 km to 20 km** and defaults to **5 km**; the chosen value is sent as `radius_km`.

Actual result	Status
Slider labelled 1 km – 20 km, default reading 5 km . Evidence: <code>screenshots/10-clinics.png</code>	 Pass

TC-CLIN-03 — Geolocation permission denied

Field	Detail
Requirement	FR-052a
Priority	Medium

Steps — Block location permission, click **Use My Location**.

Expected result — An explanatory message is shown; no API call is made; the page does not crash.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-CLIN-04 — Initial empty state

Field	Detail
Requirement	UI-5
Priority	Low

Expected result — Before any search, the page shows "Find clinics near you — Click 'Use My Location' to search for nearby healthcare facilities."

Actual result	Status
Empty state renders verbatim, with the OpenStreetMap attribution line "Powered by OpenStreetMap Data may not reflect all facilities". Evidence: <code>screenshots/10-clinics.png</code>	 Pass

TC-CLIN-05 — Out-of-range radius is rejected

Field	Detail
Requirement	FR-049
Priority	Medium
Test data	GET /clinics?lat=31.5&lon=74.3&radius_km=100

Expected result — HTTP **422** — the API constrains `radius_km` to 0.5–25.

Actual result	Status
(to be filled by tester)	 Pending

TC-CLIN-06 — Overpass mirror fallback

Field	Detail
Requirement	FR-050, NFR-Q1
Priority	Medium
Setup	Block the first mirror host at the network level

Expected result — The backend transparently retries the second mirror and still returns results; the user sees no error.

Actual result	Status
(to be filled by tester)	 Pending

11. Test Cases — Settings and Profile (TC-SET)

TC-SET-01 — Theme switching

Field	Detail
Requirement	FR-053
Priority	Medium

Steps — In Settings → Appearance, select Light, then Dark, then System.

Expected result — The interface updates immediately for each choice and the selection survives a page reload.

Actual result	Status
Settings page renders an Appearance section with theme options. Evidence: <code>screenshots/12-settings.png</code> . Switching not exercised.	 Pending

TC-SET-02 — Analysis preference toggles

Field	Detail
Requirement	FR-054, FR-055
Priority	Medium

Expected result — "Show AI detection boxes on images", "Show confidence heatmap overlay" and "Email notifications for critical findings" toggle and persist.

Actual result	Status
All three toggles render under Analysis Preferences and Notifications . Evidence: <code>screenshots/12-settings.png</code> . Persistence not exercised.	 Pending

TC-SET-03 — Preferences persist server-side

Field	Detail
Requirement	FR-056
Priority	Medium

Steps — Change a preference, sign out, sign in from a different browser.

Expected result — The preference is restored from `profiles.settings` , proving it is stored server-side and not only in local storage.

Actual result	Status
<i>(to be filled by tester)</i>	Pending

TC-SET-04 — Profile and avatar update

Field	Detail
Requirement	FR-009
Priority	Medium

Steps — On `/profile`, change the display name and upload an avatar.

Expected result — `PATCH /auth/profile` and `POST /auth/avatar` return 200; the header updates immediately; the new avatar survives a reload (cache-buster applied).

Actual result	Status
Profile page renders with the user's name and profile details. Evidence: <code>screenshots/11-profile.png</code> . Update not exercised.	 Pending

12. Test Cases — System and Infrastructure (TC-SYS)

TC-SYS-01 — Health endpoint

Field	Detail
Requirement	FR-057
Priority	High

Steps — `curl https://<backend>/health` with no authentication.

Expected result — HTTP 200 with `{"status": "ok"}`.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-SYS-02 — Root metadata endpoint

Field	Detail
Requirement	FR-057
Priority	Low

Expected result — HTTP 200 with `status` , `service` , `version: "2.1.0"` and the four model names.

Actual result	Status
<i>(to be filled by tester)</i>	Pending

TC-SYS-03 — API documentation is reachable

Field	Detail
Requirement	FR-058, NFR-Q10
Priority	Medium

Expected result — `/docs` renders Swagger UI listing all 21 endpoints; `/redoc` renders ReDoc.

Actual result	Status
<i>(to be filled by tester)</i>	Pending

TC-SYS-04 — Requests are accepted during model preloading

Field	Detail
Requirement	FR-059, NFR-P2, C-5
Priority	Critical

Steps — Restart the backend and immediately poll `GET /health` .

Expected result — `/health` responds 200 within ~5 seconds while the preload thread is still loading models; the log shows "accepting requests" before "All AI models loaded".

Actual result	Status
<i>(to be filled by tester)</i>	Pending

TC-SYS-05 — Lazy loading with `DISABLE_PRELOAD=true`

Field	Detail
Requirement	FR-060
Priority	High

Steps — Set `DISABLE_PRELOAD=true` , restart, then call `/auth/login` , `/chat` and `/diet` before any analysis.

Expected result — Non-ML endpoints work immediately; the log shows the preload skip message; the first `/analyze` call loads its model on demand and succeeds.

Actual result	Status
<i>(to be filled by tester)</i>	Pending

13. Test Cases — Security (TC-SEC)

TC-SEC-01 — Expired token is rejected

Field	Detail
Requirement	FR-007, NFR-SEC2
Priority	Critical
Setup	Set <code>JWT_EXPIRY_HOURS</code> to a very small value, or craft an expired token

Expected result — HTTP **401** "Invalid or expired token"; the frontend clears the token and redirects to login.

Actual result	Status
<i>(to be filled by tester)</i>	Pending

TC-SEC-02 — Tampered token signature is rejected

Field	Detail
Requirement	NFR-SEC2
Priority	Critical

Steps — Alter one character of a valid JWT payload and send it.

Expected result — HTTP **401**; signature verification fails; no data is returned.

Actual result	Status
<i>(to be filled by tester)</i>	Pending

TC-SEC-03 — Missing Authorization header

Field	Detail
Requirement	FR-007
Priority	Critical

Steps — Call `GET /scans` with no header.

Expected result — HTTP **401**; no scan data is leaked.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-SEC-04 — Cross-account data isolation

Field	Detail
Requirement	BR-1, NFR-SEC3, NFR-SEC4
Priority	Critical

Steps — As U2, call `GET /scans/{U1_scan_id}` and `DELETE /scans/{U1_scan_id}`.

Expected result — Both return **404**; U1's scan still exists afterwards.

Actual result	Status
<i>(to be filled by tester)</i>	 Pending

TC-SEC-05 — Row Level Security is enabled

Field	Detail
Requirement	NFR-SEC3
Priority	Critical

Steps — In the Supabase SQL editor, query `pg_tables` / policy metadata for the four tables.

Expected result — RLS is enabled on `profiles`, `scans`, `chat_sessions`, `chat_messages`, each with the documented policies.

Actual result	Status
(to be filled by tester)	 Pending

TC-SEC-06 — Default JWT secret is refused

Field	Detail
Requirement	NFR-SEC2
Priority	High
Setup	Set <code>JWT_SECRET</code> to the placeholder from <code>.env.example</code>

Expected result — The server refuses to operate with the default secret rather than starting insecurely.

Actual result	Status
(to be filled by tester)	 Pending

TC-SEC-07 — CORS blocks a disallowed origin

Field	Detail
Requirement	NFR-SEC6, CM-4
Priority	High

Steps — Send a preflight `OPTIONS` request with `Origin: https://evil.example`.

Expected result — No permissive `Access-Control-Allow-Origin` for that origin; the browser blocks the request.

Actual result	Status
(to be filled by tester)	 Pending

TC-SEC-08 — Global rate limit

Field	Detail
Requirement	NFR-SEC5
Priority	Medium

Steps — Send 201 requests to a non-AI endpoint within 60 seconds.

Expected result — The 201st returns HTTP **429**.

Actual result	Status
(to be filled by tester)	Pending

TC-SEC-09 — No secrets in the repository

Field	Detail
Requirement	NFR-SEC8
Priority	Critical

Steps — Run `git ls-files | grep -i env` and inspect every match.

Expected result — No file containing a service-role key, `OPENROUTER_API_KEY` , `HF_TOKEN` or `JWT_SECRET` is tracked. Only `.env.example` with placeholders is present.

Actual result	Status
<code>backend/.env</code> is correctly ignored. Defect: the root <code>.env</code> is tracked (contains <code>VITE_SUPABASE_URL</code> and the Supabase publishable anon key). No service-role key, LLM key or JWT secret is exposed. Logged as DEF-01 .	Fail

14. Test Cases — Performance (TC-PERF)

Warm the backend with three requests before measuring. Record the median of five runs.

TC-PERF-01 — End-to-end analysis time

Field	Detail
Requirement	NFR-P1
Target	≤ 15 s from submit to rendered result on a warm backend

Actual result	Status
(record median of 5 runs)	Pending

TC-PERF-02 — Non-AI endpoint latency

Field	Detail
Requirement	NFR-P3
Target	/scans , /stats , /auth/me respond in ≤ 1 s at p95

Actual result	Status
(record p95 over 20 calls each)	 Pending

TC-PERF-03 — Chat reply latency

Field	Detail
Requirement	NFR-P4
Target	≤ 10 s typical

Actual result	Status
(record median of 5 messages)	 Pending

TC-PERF-04 — Concurrent analyses

Field	Detail
Requirement	NFR-P7
Target	3 simultaneous analyses all complete successfully

Actual result	Status
(to be filled by tester)	 Pending

TC-PERF-05 — Event loop remains responsive during inference

Field	Detail
Requirement	NFR-P5

Field	Detail
Target	GET /health responds in ≤ 1 s while an analysis is running

Actual result	Status
(to be filled by tester)	 Pending

15. Test Cases — Usability and Compatibility (TC-UI)

TC-UI-01 — Landing page renders completely

Field	Detail
Requirement	UI-1
Priority	Medium

Expected result — Hero, Capabilities, Workflow, Report, Trust and CTA sections all render, with scroll-reveal animations completing as the user scrolls.

Actual result	Status
All six sections render at 1440×900 after scrolling; no blank regions remain. Evidence: <code>screenshots/01-landing.png</code> and section shots <code>01a – 01e</code> .	 Pass

TC-UI-02 — Mobile responsiveness

Field	Detail
Requirement	UI-1
Priority	High

Expected result — At 390×844 the layout reflows to a single column with no horizontal scrolling and no overlapping text.

Actual result	Status
Landing page renders correctly at 390×844. Evidence: <code>screenshots/14-landing-mobile.png</code> . Authenticated pages not checked at mobile width.	 Pending

TC-UI-03 — Educational-use disclaimer is always visible

Field	Detail
Requirement	NFR-S1, LR-1, UI-6
Priority	Critical

Expected result — An "Educational use only" indicator is present in the app shell on every authenticated page, and a fuller disclaimer appears on the results page.

Actual result	Status
"EDUCATIONAL USE ONLY" badge is pinned in the sidebar on every authenticated page; the results page footer reads "Educational use only. XRayVision AI is not a licensed medical device...". Evidence: <code>screenshots/05-dashboard.png</code> , <code>screenshots/13-results.png</code>	 Pass

TC-UI-04 — Three-step analysis wizard is self-explanatory

Field	Detail
Requirement	NFR-Q3
Priority	Medium

Expected result — The upload page presents numbered steps (Upload image → Analysis route → Context) and states the accepted formats and size limits on screen.

Actual result	Status
Steps 1–3 are labelled; the drop zone states "DICOM / PNG / JPG · MIN 200×200 PX · 100 KB – 20 MB"; an image-quality advisory is shown. Evidence: <code>screenshots/06-analyze-upload.png</code>	 Pass

TC-UI-05 — Urgency is not conveyed by colour alone

Field	Detail
Requirement	UI-4, NFR-Q11
Priority	High

Expected result — Every urgency badge carries a text label (`HIGH` , `CLEAR` , `MEDIUM` , ...) alongside its colour.

Actual result	Status
Badges render as coloured chips containing the words HIGH / CLEAR / MEDIUM on both the dashboard and the results page. Evidence: screenshots/05-dashboard.png , screenshots/13-results.png	✓ Pass

TC-UI-06 — Cross-browser rendering

Field	Detail
Requirement	Operating environment §2.4
Priority	Medium

Expected result — The application renders and functions in Chrome, Edge, Firefox and Safari.

Actual result	Status
Verified in Chromium 143 only. Evidence: full screenshot set. Edge, Firefox and Safari not checked.	⬜ Pending

16. Test Summary Report

16.1 Coverage by Module

Module	Cases	Verified ✓	Pending ⬜	Failed ✗
Authentication	11	1	10	0
Image Analysis	16	2	14	0
Scans and Records	12	4	8	0
Dashboard	5	4	1	0
Chatbot	8	0	8	0
Diet Planner	6	0	6	0
Clinic Locator	6	2	4	0
Settings and Profile	4	0	4	0
System	5	0	5	0
Security	9	0	8	1
Performance	5	0	5	0
Usability	6	4	2	0

Module	Cases	Verified 	Pending 	Failed 
Total	93	17	75	1

16.2 Requirements Coverage

SRS group	Requirements	Covered by
FR-001 – FR-010a	11	TC-AUTH-01 ... 11
FR-011 – FR-025a	16	TC-ANLZ-01 ... 16
FR-026 – FR-033	12	TC-SCAN-01 ... 12
FR-034 – FR-037	5	TC-DASH-01 ... 05
FR-038 – FR-043a	8	TC-CHAT-01 ... 08
FR-044 – FR-048	6	TC-DIET-01 ... 06
FR-049 – FR-052a	6	TC-CLIN-01 ... 06
FR-053 – FR-056	4	TC-SET-01 ... 04
FR-057 – FR-060	5	TC-SYS-01 ... 05
NFR-SEC1 – SEC9	9	TC-SEC-01 ... 09
NFR-P1 – P7	5	TC-PERF-01 ... 05
UI-1 – UI-7, NFR-Q3/Q11	6	TC-UI-01 ... 06

Every functional requirement has at least one corresponding test case.

16.3 Sign-off

Role	Name	Signature	Date
Tester			
Tester			
Supervisor	Maam Misbah		

17. Defect Log

ID	Test case	Severity	Description	Status	Fix
DEF-01	TC-SEC-09	Medium	The root <code>.env</code> file is tracked in git. It contains <code>VITE_API_URL</code> , <code>VITE_SUPABASE_URL</code> and the Supabase publishable anon key. No service-role key, LLM key or JWT secret is exposed, so this is a hygiene issue rather than a credential compromise — but the file is listed in <code>.gitignore</code> and should not be tracked.	Fixed	<code>.env</code> untracked with <code>git rm --cached .env</code>
DEF-02	Documentation review	Low	<code>backend/README.md</code> instructs the operator to create the <code>xray-images</code> bucket as private , but <code>supabase_client.py</code> creates it with <code>public: True</code> and serves images via <code>get_public_url()</code> . The root <code>README.md</code> correctly says public.	Fixed	Backend README corrected to say the bucket is public

ID	Test case	Severity	Description	Status	Fix
DEF-03	Medium	The fracture detector was observed emitting a non-anatomical class label (<code>vase</code>) as a low-confidence secondary finding on a wrist radiograph, alongside a correct high-confidence fracture finding from the screening classifier. Root cause identified: the bundled <code>backend/models/fracture_yolov8.pt</code> is correct — its class vocabulary is <code>Fracture / Not_Fracture</code> and contains no COCO labels. A generic COCO checkpoint, <code>backend/yolov8n.pt</code> , was also present in the repository, and its vocabulary does contain <code>vase</code> , <code>person</code> and <code>toothbrush</code> . The deployed instance was therefore running the generic checkpoint rather than the fracture model. The <code>ALLOW_GENERIC_YOLO_WEIGHTS</code> guard did not prevent this because it matches on the weight <i>filename</i> only, so a generic checkpoint reached through a misconfigured <code>YOLO_WEIGHTS_PATH</code> still loads. Impact: non-clinical labels appear in the Secondary and Borderline tiers. Confidence tiering limited the harm — the spurious labels appeared at 52.7 % and below while the correct finding held the Primary tier — but the behaviour is incorrect. Remediation: verify <code>YOLO_WEIGHTS_PATH</code> and <code>ALLOW_GENERIC_YOLO_WEIGHTS</code> on the deployed backend; strengthen the guard to inspect the loaded model's class vocabulary instead of its filename; the stray <code>yolov8n.pt</code> has been removed from version control.	Open	Fix deployed weights config; validate class vocabulary at load	
	(add further defects as they are found)				

Severity definitions — **Critical:** data loss, security breach, or a core feature unusable. **High:** a major feature fails with no workaround. **Medium:** a feature fails but a workaround exists. **Low:** cosmetic or documentation issue.

Appendix — Revision History

Version	Date	Author	Description
1.0	August 2026	Muhammad Ali Raza, Hamza Afzal	Initial test plan and 93 test case specifications; 17 verified against the live deployment

